Testing LD clusters instead of SNPs

What complexity reduction buys, and how to score it

module_sim_LDscnR

01 September 2026

Contents

1 The problem

A genome scan tests every SNP, but the object of interest is a **locus**. In linkage disequilibrium those are not the same thing, and the difference is not cosmetic: on one pooled panel of 294,609 markers, a single-SNP scan controlling FDR at 5% returned **2,467 significant SNPs, which were 132 distinct LD clusters**. The rejected SNPs were 18.7 per cluster against a panel average of 3.0, so the small-p tail was **6.3 times more redundant** than the genome as a whole.

Benjamini–Hochberg remains valid here — FDR control holds under positive regression dependence, which linkage supplies — but the quantity it controls is the false discovery proportion **among SNPs**. A reader interpreting "5% FDR over 2,467 discoveries" is being given a guarantee about a different object from the one they are counting.

Testing near-independent LD clusters makes the controlled quantity FDR **over loci**. That is an argument about what the procedure guarantees, not about which method scores better, and it holds on any dataset — with or without known causal positions.

2 Method

Markers are grouped into **stage-2 LD clusters** by the two-stage complexity chain (`ld_complexity_reduction` at $\rho = 0.5$, then `ld_prune_and_eMLG` with `min_r2_rho = 0.5`, `distance_threshold = 1e5`). Every marker belongs to exactly one cluster; nothing is discarded. Cluster sizes are strongly skewed — a median of 1 with a long tail — so most clusters are single markers and a minority are large linkage blocks.

Each cluster receives one p-value by **Simes combination over its members**, which is valid under positive regression dependence and uses all member evidence rather than a single marker. BH is then applied across clusters.

Choice	**Why**
Unit = stage-2 LD cluster	Fixed before the phenotype is seen; every marker belongs to one
Simes over members	Uses all member p-values, valid under PRDS. Outperforms every single-marker summary tested, including the LD-central representative, the highest-ld_w member and the uncorrected minimum
BH across clusters	The controlled quantity becomes FDR over loci
No ld_w threshold	ld_w builds the clusters but sets no cut. Requiring several members of a block to agree discriminates in the same direction with no free parameter

3 Scoring: do not expect one region per QTN

A causal variant sits inside one cluster, but several *neighbouring* clusters may be in linkage with it. How those are counted changes the numbers substantially, so any figure must name its convention.

Convention	**A flagged cluster counts as a true positive if...**
containment	it physically contains a driving QTN. Strictest; every LD neighbour is an error.
tagging	it reaches $r^2 \geq r^2_{\text{min}}$ with a QTN within d_{max}. Most permissive.
dedup	it is the *best* tagger of a QTN. One region per QTN; further taggers are removed rather than counted as errors. This is the `evaluate_ors` convention and the one used below.

On one panel the same Simes result scored **0.429 precision by containment, 0.643 by tagging and 0.545 by dedup**. The three answer different questions; none is wrong.

The dedup convention is the informative one because it separates two kinds of error that behave quite differently: a **satellite** is a genuine LD neighbour of a locus already found, whereas a cluster that tags nothing is simply wrong.

4 Result

4.1 Where the flagged regions go

Table 1: Median flagged stage-2 clusters per panel, decomposed. 80 panels: 4 simulation cells x 2 background-selection arms x 10 environments, each pooling 10 map sets into a genome-wide panel of about 90,000 clusters.

an	flagged	contains	satellite	tags nothing
single SNP	56	6.5	2.5	44
representative	5	2.0	0.0	3
Simes	10	3.0	0.0	4
eMLG	13	3.0	0.0	7

The single-SNP scan's extra output is overwhelmingly noise, not near misses. Its median panel flags 56 clusters of which 44 tag no QTN anywhere, against 2 satellites. Cluster-level testing flags 10 and recovers 3 QTN-bearing clusters.

Satellites are a minor share of the errors in *every* analysis. Merging adjacent flagged clusters through a second clustering step therefore has a low ceiling: it never increases the region count (0 wins in 320 paired comparisons) and improves precision in 166 of 168 non-tied cases, but by a median of only 0.002.

4.2 Precision and recall

Table 2: Median over 80 panels, dedup convention.

an	precision	recall	PR
single SNP	0.102	0.351	0.0386
representative	0.191	0.114	0.0268
Simes	0.229	0.169	0.0464
eMLG	0.203	0.179	0.0271

Cell	Simes wins	losses	ties	sign test
`V0.5_c1`	19	0	0	4e-06
`V0.5_c2`	8	10	1	0.815
`V1_c1.5`	7	5	2	0.774
`V2_c1`	11	7	0	0.481
pooled	45	22	3	0.007

5 What this means for a call on real data

The practical upshot is one sentence, and it follows from the FDP argument rather than from any performance number: **a call from the clustered analysis is about twice as likely to be real as one from a single-SNP scan**, and it is more likely to be real *because the error rate is finally about the thing being counted.*

Engine	Arm	single SNP	Simes	ratio
EMMAX	`nobgs`	0.101	0.254	2.5×
EMMAX	`bgs`	0.111	0.194	1.7×
LFMM	`nobgs`	0.079	0.164	2.1×
LFMM	`bgs`	0.121	0.252	2.1×

Precision of flagged clusters, medians over 40 panels per cell. The **ratio** is the transferable part; the absolute values are properties of this signal-to-noise regime.

This is not extra caution, and both analyses control FDR at 5%. What differs is what the guarantee is *about.* A single-SNP scan controls the false discovery proportion among SNPs while being judged against a locus-level truth; clustering aligns the two. The higher precision is that alignment, not conservatism added on top.

Two limits on how far to take it.

It is comparative, not absolute. Precision is about 0.25 here under the strict containment convention — three calls in four still do not contain a QTN — and 0.64 under tagging. A real dataset's value is not predicted by either.

And it is not free. The same property costs recall, which is what Section~6 prices: below $\beta \approx 2$ the trade is worth making, above it is not. *"Fewer false positives"* and *"more missed loci"* are the same fact stated twice.

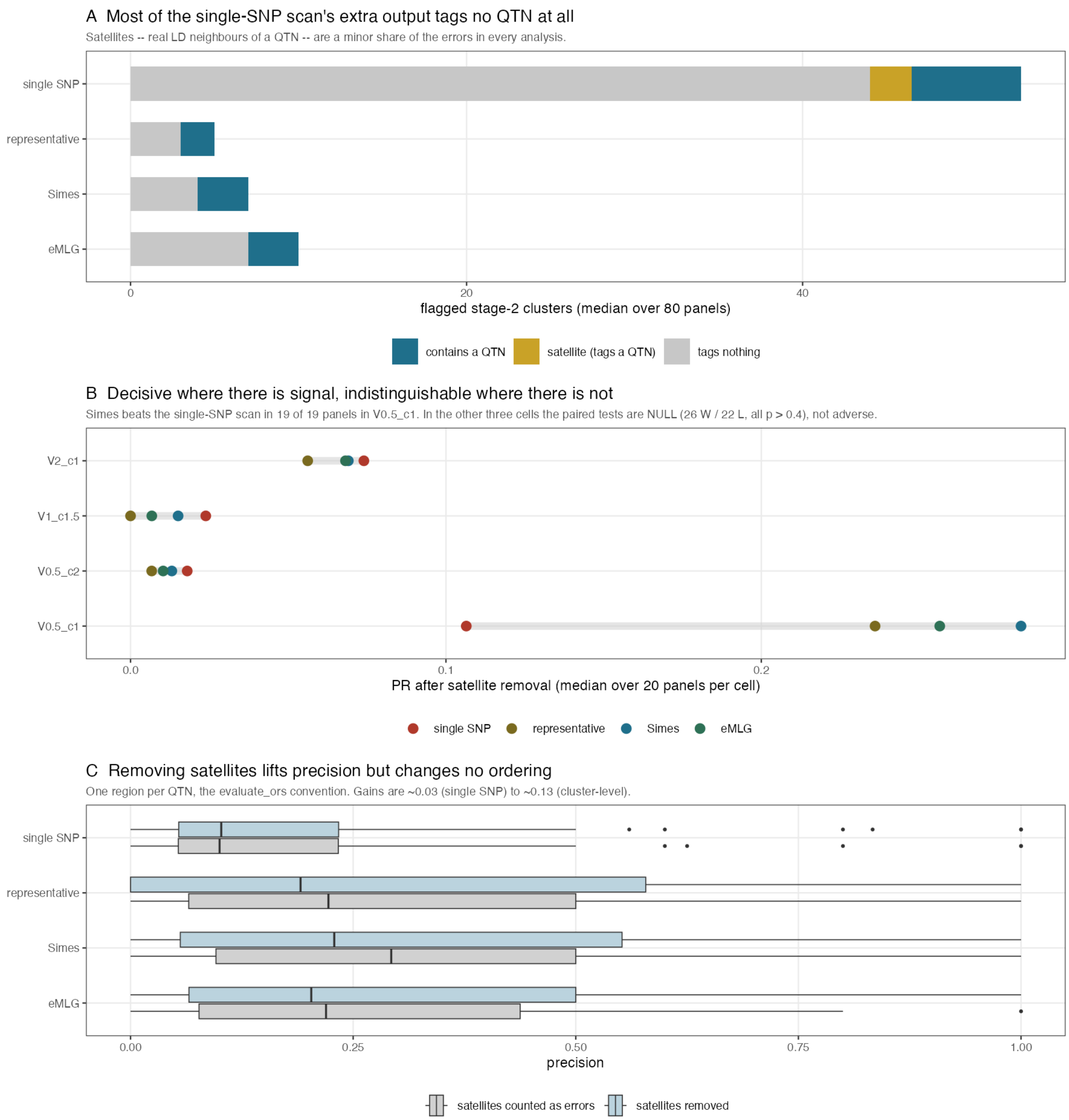

Figure 1: Where flagged clusters go, how the advantage varies by simulation cell, and what satellite removal changes.

5.1 Replication with a second association method

Everything above uses EMMAX. The cluster partition is built from genotypes and never sees a p-value, so any method supplying per-marker p-values substitutes by changing one column. Repeating the whole comparison with **LFMM**:

Engine	Analysis	flagged	contains a QTN	tags nothing
EMMAX	single SNP	67.5	7.0	51.0
EMMAX	Simes	10.5	3.0	7.0
LFMM	single SNP	94.0	8.0	75.5
LFMM	Simes	25.0	5.5	15.0

`nobgs` arm, medians. **80% of the single-SNP scan's flags tag no QTN under LFMM**, against 60% for Simes — the same shape as EMMAX at larger magnitude. And the β crossover sits at ≈ 2 under both engines, so it is a property of the comparison rather than of EMMAX.

6 The metric is a claim, not a convention

Clustering and the single-SNP scan sit at **different operating points**, so which is "better" is decided by how precision and recall are weighted — and that choice is an assumption about the use, not a neutral convention.

	`nobgs`		`bgs`	
Analysis	precision	recall	precision	recall
single SNP	0.101	**0.372**	0.111	**0.348**
Simes over cluster members	**0.254**	0.155	**0.194**	0.177
eMLG	0.225	0.165	0.200	0.179
LD-central representative	0.262	0.150	0.125	0.095

Simes has **higher precision in 62 of 71 non-tied panels** — 34 W / 3 L in `nobgs` ($p = 1.2 \times 10^{-7}$) and 28 W / 6 L in `bgs` ($p = 2.0 \times 10^{-4}$) — and lower recall. The single-SNP scan finds more loci and buries them: a median of 51 of its 67 flagged clusters tag no QTN at all, against 7 of 10 for Simes.

6.1 Where the comparison turns

$F_\beta = (1 + \beta^2)PR/(\beta^2 P + R)$ weights recall β times as heavily as precision.

β	median F (single SNP)	median F (Simes)	paired over 80 panels
0.25	0.107	**0.229**	60 W / 12 L, $p = 8 \times 10^{-9}$
0.50	0.120	**0.185**	56 W / 16 L, $p = 2 \times 10^{-6}$
1.00	0.152	**0.181**	48 W / 24 L, $p = 0.006$
1.50	**0.188**	0.153	43 W / 29 L, $p = 0.12$
2.00	**0.214**	0.147	34 W / 38 L, $p = 0.72$
3.00	**0.270**	0.169	20 W / 52 L, $p = 2 \times 10^{-4}$
4.00	**0.319**	0.174	15 W / 57 L, $p = 7 \times 10^{-7}$

The crossover lies between $\beta = 1.5$ and 2. Weight recall no more than about 1.5 times precision and clustering wins; weight it twice or more and the single-SNP scan wins. At the balanced F_1 clustering wins ($p = 0.006$).

The $P \times R$ product used elsewhere in this module sits near $\beta = 1$, so the results reported there are conditional on that weighting. **This is stated rather than buried:** a single performance number cannot settle this comparison, and any claim that one analysis is better is a claim about β.

Which regime applies depends on the use. For a scan whose output is a candidate list for experimental follow-up, each false positive costs real work and $\beta < 1$ is defensible — clustering wins there decisively, 60 W / 12 L at

$\beta = 0.25$. For a discovery scan feeding a meta-analysis, where a missed locus is the expensive error, $\beta > 2$ is defensible and the single-SNP scan wins. Neither is the default.

7 Scope, and what it means for real data

Cluster-level testing wins decisively where there is signal to aggregate, and is **indistinguishable** where there is not. It is not worse anywhere. In the low-signal cells the paired comparisons are null, not adverse — several panels are ties in which neither analysis recovers anything, so those panels carry no information about which method is better.

Two consequences for an empirical analysis, where causal positions are unknown:

The core argument needs no truth. Testing near-independent units makes the reported FDR a statement about loci. That follows from what BH controls, not from any performance comparison, so it transfers directly.

The performance numbers do not transfer. Precision and recall here are simulation quantities. What can be checked on real data without truth is calibration under a structure-preserving permutation, replication of the same clusters across environments or subsets, and stability under resampling.

8 Provenance

Simulations: `regen_sim_data_bgs5`, 4 cells $\times$ 10 environments $\times$ 2 BGS arms $\times$ 10 map sets. Association: EMMAX per marker against a GRM built from the pruned markers, so clustering conditions the null rather than the test. Truth: driving QTN, defined as MAF > 0.1 with within-chromosome $p_{Va} \geq 0.05$. Scripts: `snp_vs_cluster_dedup.R`, `cluster_summary_test.R`, `plot_clustering_gain.R`.

Every comparison is **paired within panel** and reported with wins, losses, ties and a sign test. Differences of means are not used.

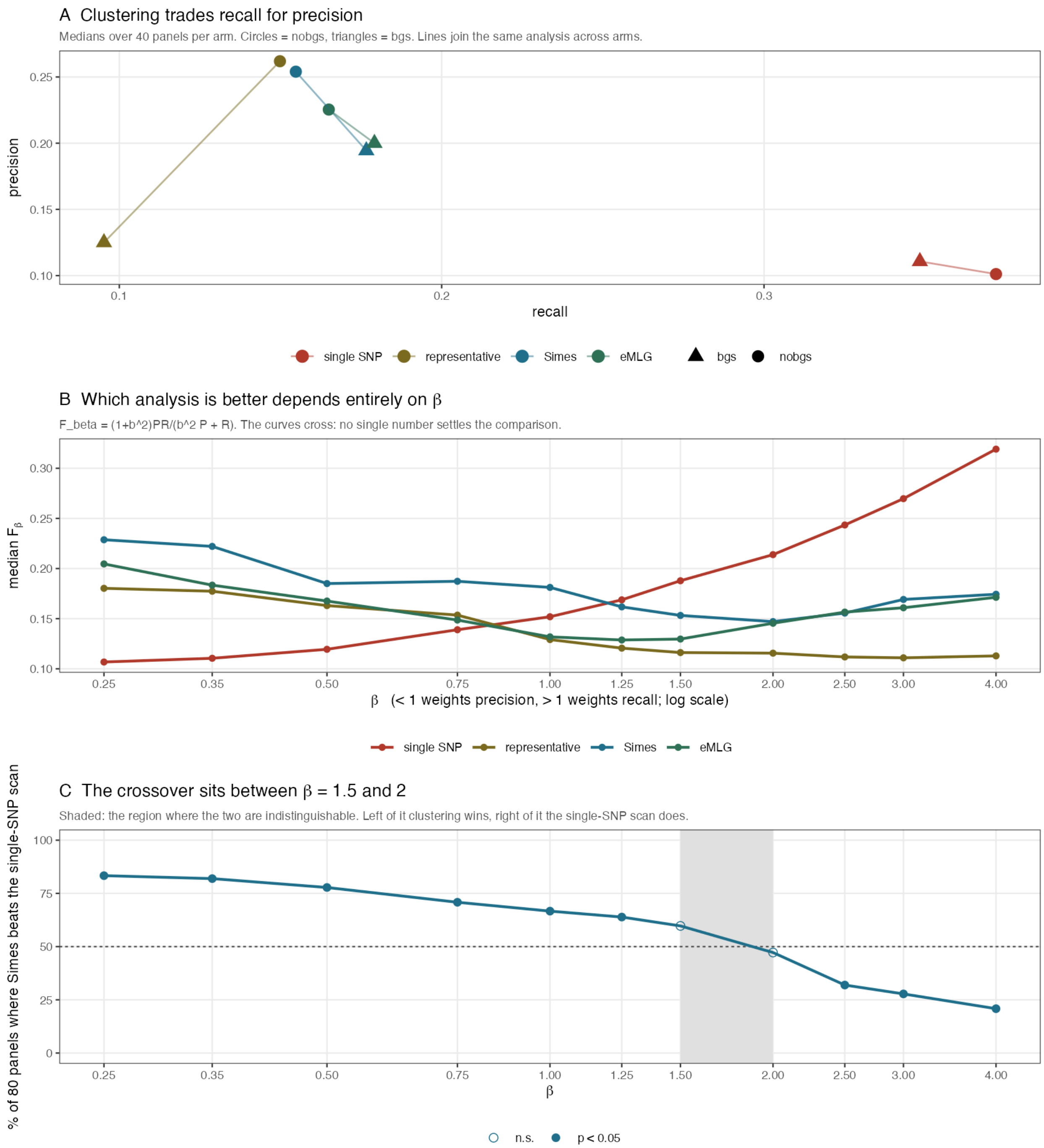

Figure 2: The trade-off. A: the two analyses occupy different operating points. B: median F_β curves cross, so no single number settles the comparison. C: the paired win rate against β, with the indistinguishable region shaded.